

Derbyshire Shared Care Pathology Guidelines

Secondary Hypertension

Purpose of Guideline

This guideline covers the investigation and referral criteria of patients with suspected secondary causes of hypertension. Appropriate tests that may be carried out in primary care prior to referral are outlined. (The clinical management of patients with primary hypertension is covered by the NICE guideline CG127).

Definition

Hypertension is defined as persistent raised blood pressure and is classified according to level of blood pressure:

- Stage 1: Clinic BP above 140/90 mmHg and subsequent average of home monitoring of 135/85 or higher
- Stage 2: Clinic BP above 160/100 mmHg and subsequent average of home monitoring of 150/95 or higher
- Severe: Systolic BP 180mmHg or higher or diastolic 110 mmHg or higher

**Consider Urgent Hospital Admission for Patients with Accelerated Hypertension
Systolic BP>180 and/or Diastolic >110
AND
Signs of Papilloedema and/or Retinal Haemorrhage**

When should secondary causes of hypertension be suspected?

When 1 or more of the following is present:

- Blood pressure remains uncontrolled with the optimal or maximum tolerated doses of multiple antihypertensive drugs
- Patients aged <40 years with evidence of target organ damage, cardiovascular disease, renal disease or diabetes
- Hypokalaemia

Background

Secondary causes of hypertension include:

COMMON

- Renal insufficiency

RARE

- Mineralocorticoid excess
 - Hyperaldosteronism (Conn's syndrome)
 - Steroid pathway defects e.g. 11 β OHase deficiency CAH (rare)
 - Excess liquorice ingestion (rare)
- Glucocorticoid excess
 - Cushing's syndrome/disease
 - Corticosteroid treatment
 - 11 β hydroxysteroid dehydrogenase deficiency (rare)
- Pheochromocytoma
- Acromegaly

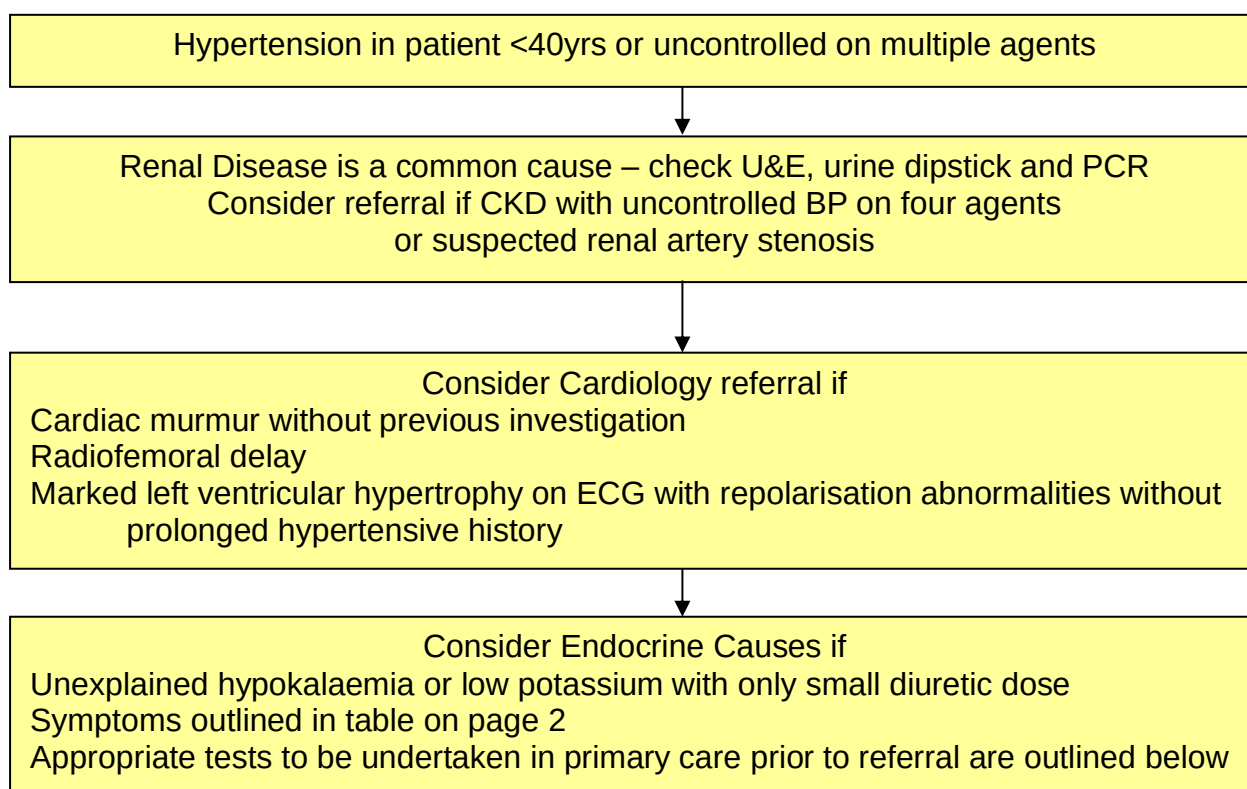
Signs and symptoms of secondary causes of hypertension

Signs/symptoms	Possible cause	Prevalence
Low Potassium (excluding diuretic induced hypokalaemia) NOTE: 50% of patients with Conn's do not have hypokalaemia. Low potassium brought on by a small dose of diuretic may be a clue	Primary Hyperaldosteronism (Including Conn's) Secondary Hyperaldosteronism (e.g. Renal Artery Stenosis) Apparent Mineralocorticoid Excess (e.g. liquorice ingestion)	Conflicting data; 0.5 – 5.0% of hypertension cases
Cushingoid appearance + Oligomenorrhoea Easy bruising	Cushing's Glucocorticoid treatment	Cushing's up to 13 per 10 ⁶ Steroid excess unknown
Palpitations, sweats, postural hypotension, anxiety	Pheochromocytoma	0.05 – 0.2% of hypertensive population
Acromegalic appearance	Acromegaly	3-4 per million per year of general population. Hypertension seen in 35% of cases ^b .

What to do next

If not already undertaken recently all patients should have an Initial Biochemical screen:

- UE's
- LFT's
- Calcium
- Fasting Lipids
- Fasting Glucose
- Magnesium
- Random urine
 - Dipstick for blood, protein and glucose
 - Send to the lab for protein/creatinine ratio and UE (if patient hypokalaemic)



Suspected primary and/or secondary hyperaldosteronism

The accurate measurement of Plasma Renin and Aldosterone is important in the investigation of suspected primary hyperaldosteronism such as Conn's syndrome. Pre-analytical variables (e.g. drugs, posture, sodium intake) will affect the results and interpretation and as such it is recommended that patients with unexplained hypokalaemia and hypertension should be referred to a Consultant Endocrinologist for further assessment.

A guideline - First Line investigation of suspected Hyperaldosteronism for Primary Care- is available on the shared care pathology website in the Endocrinology section, however note that samples for renin must be processed within 15 minutes of venepuncture.

Suspected Cushing's Syndrome

The recommended screening test for Cushing's syndrome is the **low dose overnight dexamethasone suppression test**. Random cortisol levels are not useful in this setting as normal levels do not exclude adrenal hyperfunction. This test can be performed in primary care or alternatively patients can be referred to a Consultant Endocrinologist for investigation.

Supply the patient with a 1 mg dose of dexamethasone (2 x 500 mcg tablets). Instruct the patient to take both tablets together, between 23.00 and midnight, on any night Sunday – Thursday.

Give the patient a completed request form, for a post-Dexamethasone cortisol and instruct them to attend for a blood test the following morning between 09:00 and 09:30 at a suitably convenient phlebotomy location.

Always clearly label the request form with Dexamethasone Suppression Test

INTERPRETATION:

Normal suppression is indicated by a 09:00 serum cortisol <50 nmol/L.

NOTE that failure of cortisol suppression may indicate cortisol excess but also sometimes occurs in:

- Severe endogenous depression
- Alcoholism (pseudo-Cushing's syndrome)
- Severe stressful illness/infection
- Hepatic enzyme inducing drugs (phenytoin, phenobarbitone, rifampicin etc)
- Oestrogen therapy
- Failure to take dexamethasone properly
- Glucocorticoid resistance syndrome
- Obesity
- Renal failure
- Steroid containing skin creams
- Beconase eye drops
- Other steroid containing OTC medication

It is recommended that all patients with abnormal Dexamethasone Suppression test results are referred to a Consultant Endocrinologist to establish a definitive diagnosis and establish the correct treatment pathway

A guideline - Overnight Dexamethasone Suppression Test for Primary Care - is available on the shared care pathology website in the Endocrinology section.

Suspected Pheochromocytoma

For screening for suspected pheochromocytoma in the GP population 24 hour urinary metanephrines should be requested. Metanephrines are metabolites of catecholamines and their analysis has superseded urine catecholamines for the investigation of possible pheochromocytoma due to increased sensitivity.

Ideally 3 consecutive 24 hour collections should be performed as the release of catecholamines and metabolites from such tumours can be episodic leading to falsely normal levels during asymptomatic periods.

24 hour urine collection bottles containing acid are required and can be ordered via pathology (Pathology Supplies: Tel 01332 785581, pathsupplies@derbyhospitals.nhs.uk) A patient information sheet with instructions for performing an accurate 24 hour urine collection is supplied with the container.

Paracetamol should be avoided for 5 days prior to commencing sample collection. Many other medications can cause false increases in catecholamine results; it is **essential to provide a full medication list with all requests** to allow appropriate interpretation.

Medication	Effect
Phenoxybenzamine	Increased urine normetanephrine and noradrenaline
Tricyclics	Increased urine normetanephrine and noradrenaline
Atenolol Metoprolol Propanolol Labetolol	Increased urine normetanephrine and metanephrine
Sulphosalazine	Increased urine normetanephrine
Levo dopa Methyl dopa	Increased urinary 3-methoxy tyramine

INTERPRETATION

Analyte	Normotensive range nmol/24hr
Normetanephrine	0 – 4400
Metanephrine	0 – 2000
3-Methoxy tyramine	0 – 2300

Note: 3-Methoxytyramine is a metabolite of dopamine and is a marker of dopamine secretion

All results are reviewed and have appropriate interpretative comments added by the duty biochemist. Borderline increases may be due to medication effects or stress but results strongly suggestive of a possible pheochromocytoma will be telephoned to the requesting source.

It is recommended that all patients with catecholamine metabolites suggestive of Pheochromocytoma are referred urgently to a Consultant Endocrinologist to establish a definitive diagnosis and establish the correct treatment pathway.

First line investigation of suspected Acromegaly

It is recommended that all patients with clinical symptoms of Acromegaly be referred to a Consultant Endocrinologist to establish a definitive diagnosis and establish the correct treatment pathway. Prior to seeing an Endocrinologist the following baseline pituitary hormone blood tests should be performed on a 9 am sample.

- LH & FSH
- Full thyroid profile (TSH, FT4 & FT3)
- Prolactin
- Testosterone (if male)
- Oestradiol (if female)
- Cortisol
- Growth Hormone & IgF1

Contacts

Duty Biochemist	01332 789383 (8 am to 7 pm, Mon – Fri)
Endocrinology Advice	07879 115507 (9 am – 5 pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131(24/7)
A&E, MAU and ACC	Via RDH switchboard, 01332 340131(24/7)
OR	
ED Nurse in Charge	07799 337674
MAU Nurse in Charge	07917 650751
Pathology Supplies	pathsupplies@derbyhospitals.nhs.uk 01332 785581

References

- a NICE clinical guideline CG127. Hypertension. Clinical management of hypertension in adults (last updated Nov 2016)
- b F.D. Rosenthal, S. Roy. Hypertension and hyperparathyroidism. *BMJ* 1972; 4;396-397
- c Bondanelli M, Ambrosio MR and Uberti E. Pathogenesis and prevalence of hypertension in acromegaly. *Pituitary* 2001, 4; 239-249
- d Lenders et al. Biochemical investigation of pheochromocytoma. *JAMA* 2002;287;1427-1434

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